



Student Name: HEWP THOS LOPS TRIA AJIA CHEY

Test Name: MATHEMATICAL METHODS

Date: _____

MATHEMATICS TEST

UNIT 2

Test theme: SPIDERMAN

Reading time: 5 minutes

Writing time: 50 minutes

<i>Number of questions</i>	<i>Number of questions to be answered</i>	<i>Number of marks</i>
3	3	32

- Students are permitted to bring into the test room: pens, pencils, highlighters, erasers, sharpeners, rulers.
- Students are NOT permitted to bring into the examination: notes of any kind, blank sheets of paper, white out liquid/tape or a calculator of any type.

Materials supplied

- Question and answer book.
- Working space is provided throughout the book.

Instructions

- Write your name in the space provided above on this page.
- All written responses must be in English or Romanian.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the test room.

Instructions

Answer **all** questions in the spaces provided

In all questions where a numerical answer is required, an exact value must be given unless otherwise specified.

In questions where more than one mark is available, appropriate working **must** be shown.

Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.

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*The Green Goblin has hijacked the Oscorp mainframe, locking Peter Parker's web-shooters with a complex encryption function. To regain control and stop the Goblin from gliding over the Brooklyn Bridge, **Spider-Man** must analyze the rate of change of the encryption's energy levels. If Peter can find the stationary points and sketch the energy curve, he can override the system. Use your knowledge of derivatives to help Spidey save New York!*

Question 1 (11 marks)

The Green Goblinc attacks with her first function,

$$f: [-2, \infty) \rightarrow \mathbb{R}, f(x) = (x + 1)^6 + (2x^3 + 6x^2 + 6x + 2) + 92 + 37$$

- a. Express $f(x)$ in the form of $b(x + a)^6 + c(x + a)^3 + d$ where $a, b, c, d \in \mathbb{Z}$

1 mark

- b. Hence or otherwise, find the rule for derivative of $f(x)$ in the form of $b(x + a)^c((x + a)^d + 1)$ where $a, b, c, d \in \mathbb{Z}$ and state the domain.

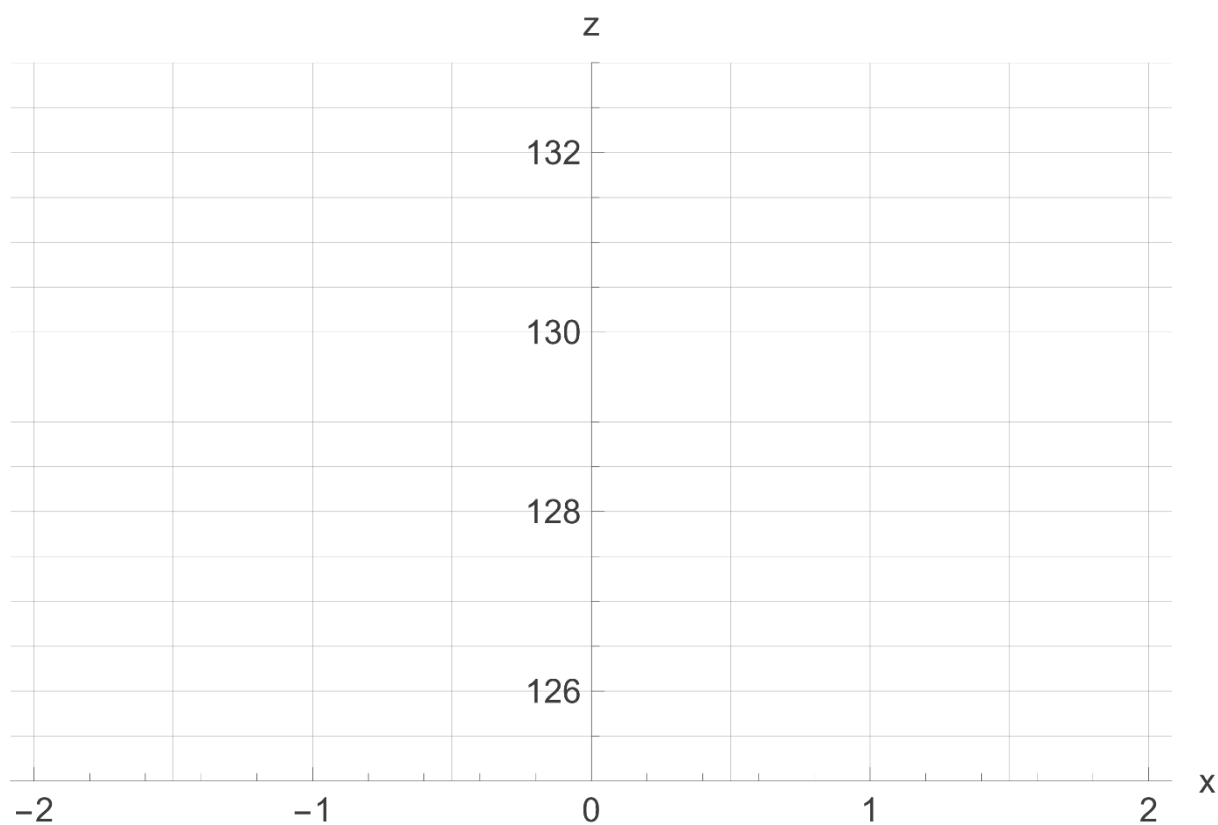
3 marks

c. Hence, find the coordinate(s) of the stationary point(s) of $f(x)$

3 marks

d. Sketch the graph of $z = f(x)$ on the axes below. Label all axis intercept(s), stationary point(s) and endpoint(s) with their coordinates.

2 marks



e. State the transformation(s) applied to $z = f(x)$ in order to transform it into $z = 2f\left(2x - \frac{1}{3}\right) - 129$. Do not use the terms up, down, left or right.

2 marks

Question 2 (9 marks)

Spider-Man is impressed, you full-marked the entire question. She begins to slowly retreat and attack with her dual-functions:

$$g: R \rightarrow R, g(t) = 2t^4 + t^3 - 5t^2 - t + 3$$

$$h: R \rightarrow R, h(w) = w^3 - w$$

a. Show that the equation of the tangent to $g(t)$ at $t = \frac{1}{2}$ is $l_1 = -\frac{17}{4}t + \frac{29}{8}$

2 marks

b. Hence, state the area bound by the vertical axis, horizontal axis and l_1

1 mark

- c. Let $l_2(w)$ be the line tangent to $h(w)$ at the point $w = -\frac{1}{2}$. State the rule of $l_2(w)$.

2 marks

- d. Let $l_3(w) = l_2(kw)$, $k \in \mathbb{R}$. Find the value(s) of k such that $l_3(w)$ makes an angle between 30 degrees and 60 degrees with the positive direction of the w -axis.

3 marks

- e. Let $h_2(w) = h(w) + p$, $p \in \mathbb{R}$. Find the value(s) of p such that $h_2(w)$ and $l_2(w)$ have only 1 point of intersection.

1 mark

Question 3 (12 marks)

The Green Goblin attempts to escape. Her position can be modelled as a function of time (seconds) in the piecewise function $p(t)$. Where p represents the position away from the point of escape in meters, and t represents time in seconds. The Green Goblin will successfully escape if she reaches the point of escape ($p=0$ meters) at or within 5 seconds.

$$p(t) = \begin{cases} 5 - \frac{2}{7}t, & 0 \leq t < \frac{7}{2} \\ \frac{1}{(t-3)^2}, & \frac{7}{2} \leq t \leq 4 \\ 6 - \frac{7}{4}t + \frac{t^2}{8}, & 4 < t \leq 6 \end{cases}$$

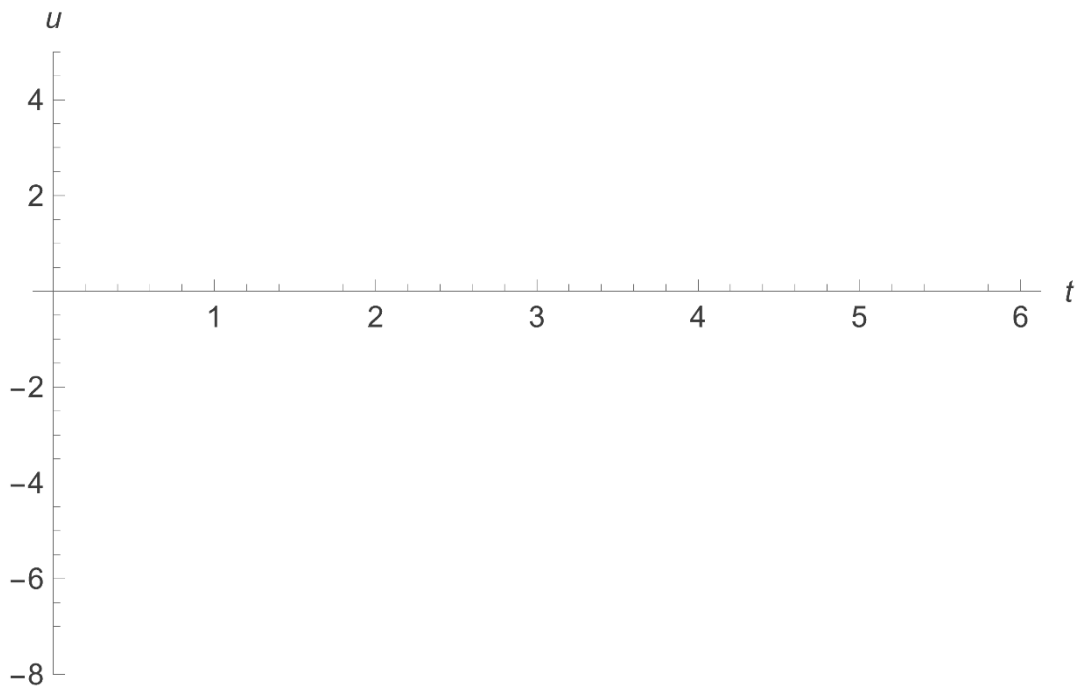
- a. Given that $p(t)$ continuous across its entire domain, is $p(t)$ differentiable across its entire domain? Justify your answer with **all** relevant calculations and statements.

2 marks

- b. It is known that the rate of change of position is velocity. Find the function for the velocity of The Green Goblin, $v(t)$, expressing it as a piecewise function.

1 mark

- c. Sketch the graph of $u = p(t)$ and $u = v(t)$ on the axes below. Label all axis intercept(s), points of discontinuity and endpoint(s) with their coordinates and graphs with their functions.

3 marks

- d. Determine the time after starting to escape where The Green Goblin's speed is at its greatest. State the speed. (Note: speed does not have direction unlike velocity)

2 marks

- e. It is known from the graph in part c that The Green Goblin does not escape in time. Determine the transformation(s) applied to the last sub-function of the piecewise, $k(t) = 6 - \frac{7}{4}t + \frac{t^2}{8}$, such that she will escape at the last possible moment whilst also maintaining that the function is continuous across its entire domain.

1 mark

- f. i) The Green Goblin realises that the middle sub-function of the piecewise ($\frac{1}{(t-3)^2}$, $\frac{7}{2} \leq t \leq 4$) is too steep, and she is unable to run that fast. She removes the middle sub-function entirely, and wishes to translate the last sub-function, $j(t) = 6 - \frac{7}{4}t + \frac{t^2}{8}$ such that it shifts horizontally to maintain the continuity at $t = \frac{7}{2}$. Assume that j takes over all of the removed domain. Find the transformation(s) applied to $j(t)$. (*part e does not affect this question*)

2 marks

- ii) Hence, determine whether The Green Goblin escapes in time under the transformed $j(t)$. Justify why or why not.

1 mark

END OF QUESTION AND ANSWER BOOK